Deinterleaving of Mixtures of Renewal Processes

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***Abstract*—This paper proposes a new method for impulsive source separation (deinterleaving) in multi-input single-output systems based purely on impulse spacing. The impulse spacing for each source is modeled as a renewal process. To classify sources, an algorithm is developed to maximize the likelihood of impulse assignment based on the impulse spacing distributions. A theoretical lower bound on performance is derived and the algorithm is shown to get reasonably close to the bound. The algorithm is extended for use when the impulse spacing distribution parameters are unknown. Performance is evaluated on multiple application examples.**

***Index Terms*—Deinterleaving, click trains, jitttered PRI radar, source separation, renewal process, marine mammals, impulsive sources, multi-input single-output.**

I. Introduction

We consider the following problem. A source $k \in \{1, \ldots, K\}$ generates events at times $\tau_{k,i} \in [0, T]$; the events could be transmission of packets or emission of an impulsive signal. An observer is given the total set of impulses $\{\tau_{k,i}\}$ without any labels. Can the observer determine which event time belongs to which source solely from the timing information?

This problem is motivated by various applications, including:

- *Computer networks:* Can the timing of packets reveal who transmitted the packets?
- *Marine mammals:* Can marine mammal echolocation clicks trains [1], [2] be separated based on their timing alone?
- *Radar and sonar:* Is it possible to separate (deinterleave [3]–[7]) multiple pulsed radar signals recorded passively using only pulse time of arrival?
- *Heart beats:* In a mixture of heartbeats, e.g., in fetal ECG [8], is it possible to say which heartbeat belongs to the mother and which to each fetus (with multiple pregnancy) based solely on timing?
- *Nervous system:* In recordings of neural spike trains, can spike trains from different sources be separated [9]?

In many of these applications, there are characteristics in addition to timing (such as impulse amplitude and shape) that can be used to help separate events/impulses. Nevertheless, there is value to fully understanding the fundamental timing separation problem by itself. For example, privacy/security concerns or limitations on transmission bandwidth may yield situations with timing information only. In another example, separating based on timing alone could facilitate an initial analysis in a case where secondary characteristics are unknown apriori. Even in cases with obvious/known secondary characteristics, our approach is classical (e.g., information theory [10]): A practical problem motivates the set-up of a simple theoretical/mathematical model which is rigorously explored and understood, and is subsequently expanded/modified for practical application. Here, our pure timing separation algorithms can be extended in the future to include secondary characteristics and to make them applicable in real-life scenarios.

Separation of impulsive sequences is not a new problem in itself. For example, several approaches appear in prior literature to separate marine mammal click trains from recordings on single sensors. [11] used click cross-correlation to establish relationships between clicks and to separate click trains. [12] separated sperm whale click trains based on correlation between clicks, click structure detail (inter-pulse-intervals), and frequency characteristics (but ultimately relied on time delay estimates between two hydrophones to separate click trains). [13] used a cluster analysis approach on features extracted from detected clicks; [14] sequentially matched the frequency spectra of successive clicks; and [15] separated click trains based on spectral dissimilarity. These methods rely on the slowly-varying nature of click pulse shapes within a click train. Only a few click train separation methods have used timing information. [16] tracked slow variations in click amplitude and inter-click-intervals (ICI) to separate two click trains. Baggenstoss' algorithms [17]–[19] define statistical measures of click similarity based on features, including ICI, extracted from click pairs and that maximize a global measure of similarity between associated clicks. As does our paper, [20] analyzed click trains based on timing *only*: It used the pulse repetition interval (PRI) transform of [21] to estimate ICIs and extended this to time varying ICIs. However, [20] did not separate (classify) each pulse.

Deinterleaving of radar signal based on pulse time of arrival only has also been considered before, e.g., [4]–[7], [22]–[25]. Radar signals usually are either periodic, staggered, or jittered.

Manuscript received February 1, 2018; revised September 14, 2018; accepted November 21, 2018. Date of publication December 10, 2018; date of current version January 4, 2019. The associate editor coordinating the review of this manuscript and approving it for publication was Dr. Joao Xavier. This work was supported in part by the NSF Grant 1017775, in part by the ONR Grant N000141612598, and in part by Shenzhen Peacock Plan under Grant KQTD2015033114415450. This paper was presented in part at the IEEE DSP Workshop, Napa Valley, CA, USA, 2013. *(Corresponding author: Anders Høst-Madsen.)*

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Digital Object Identifier 10.1109/TSP.2018.2886149

In jittered PRI[1] radar a random time is added to each PRI by the transmitter, which makes the signal a renewal process (see Section II below). Therefore, our results for renewal processes can be directly applied to deinterleaving jittered PRI radar signals. Classification based on timing alone has been considered before in [6], [7]. [6] considered noisy time of arrival and [7] used a random walk model for PRI, while [21] specifically considered jittered PRI, but only to estimate the PRIs. Other papers [22]–[25] develop heuristic methods for deinterleaving jittered PRI. The algorithms developed in our paper present optimum solutions to this problem. We compare our methods with [22]–[25] in Section VI-C1.

This manuscript develops algorithms specifically for separation (deinterleaving) of renewal processes based on timing alone. We believe that this is a new problem.

II. System Model and Problem Description

We consider a system model with K sources that each generate events or emit impulse-like signals. Each source k transmits a sequence of impulses at times $\tau_{k,i}$. We assume that the support of the impulse response is much shorter than the pulse spacing, so there is no overlap. To be precise: We assume that the probability of overlap is low enough that it can be ignored. Without loss of generality, we assume that the first impulse time occurs at $t = 0$.

We want to decide which pulse belongs to which source. Our aim is to classify based only on the (unclassified) impulse times $\{\tau_i\}$. We assume the impulses have been found by some detection algorithm with low error probability.

To be able to classify the $\{\tau_i\}$ based only on timing information, we assume the following:

- The interimpulse spacing $\tau_{k,i} - \tau_{k,i-1} > 0$ is distributed according to some parametric prior distribution $\mathcal{T}_k(t)$ whose parameters may be known or unknown.
- The interimpulse spacings $\tau_{k,i} - \tau_{k,i-1}$ and $\tau_{l,j} - \tau_{l,j-1}$ are independent for $k \neq l$ or $i \neq j$.

With these assumptions, the set of event times for a single source constitutes a renewal or counting process [26, Section 8.3]; the most well-know example is the Poisson process. Renewal processes are widely used to model many processes [26, Section 8.3], which is one reason to use this model. In particular, it is probably a good model for biological signals. At a purely speculative level, biological systems likely do not have an internal *metronome* to which they tie event (e.g., heartbeat, click) generation; rather they have a *timer* to count down to the next event. To generate nearly periodic signals, the timing distribution can then be made very narrow around a mean. The timing distribution might not be independent, but the first order approximation is usually to assume independence, which results in a renewal process. While a good model for click trains is not known, heartbeat is often modeled in this way [27].

For many applications we might not know the *type* of prior distribution $\mathcal{T}_k(t)$. Without further knowledge, a truncated Gaussian seems a reasonable assumption. One can argue for the truncated Gaussian distribution in terms of central limit theory or maximum entropy: given mean μ and variance σ^2, the positive distribution with largest entropy is exactly a truncated Gaussian distribution [28]. It is necessary to truncate the distribution at 0 to enforce the constraint of non-negative impulse spacings. We denote a Gaussian distribution truncated at 0 with mean μ and variance σ^2 by $\mathcal{N}_0(\mu, \sigma^2)$; the probability density function is $f(x) = \frac{1}{\Phi(\frac{\mu}{\sigma})\sqrt{2\pi\sigma^2}} \exp(-\frac{(x-\mu)^2}{2\sigma^2})$ for $x > 0$, where Φ is the normal cumulative distribution function [26]. In many cases we consider, $\mu \gg 0$ and $\mu \gg \sigma$. Therefore the correction $\Phi(\frac{\mu}{\sigma})$ due to truncation is close to one and can be ignored in many calculations.

We assess the performance of our algorithms by the error probability, that is the probability that an impulse is wrongly classified. Let $D_i \in \{1, \ldots K\}$ denote the source that originated the impulse at time τ_i, and $\hat{D}_i$ an estimate. We then define the error probability for source k as $P_{e,k} = \lim_{T\to\infty} P(\hat{D}_j \neq D_j | D_j = k)$ and the total error probability as

$$P_e = \lim_{T\to\infty} P(\hat{D}_j \neq D_j). \tag{1}$$

III. Timing-Based Separation With Known Parameters

In the basic approach, we assume that the number of sources and prior distributions are completely known. The problem is therefore to classify the impulses based on the impulse times $\{\tau_i\}$ only using the known prior distributions $\mathcal{T}_k(t)$. We consider the maximum likelihood (ML) solution[2]

$$\max_{\hat{\mathbf{D}}} L; \quad L = \sum_{k=1}^{K} \sum_{i} \log \mathcal{T}_k(\tau_{S_k(i)} - \tau_{S_k(i-1)}) \tag{2}$$

with respect to $S_k(i)$. Here $S_k(i)$ denotes the i-th impulse *assigned* to the k-th source, and is a function of $\hat{\mathbf{D}} = (D_1, D_2, \ldots)$. For example, if for two sources $\hat{\mathbf{D}} = (1, 1, 2, 1, 2, 2)$ then $S_1 = (1, 2, 4)$ and $S_2 = (3, 5, 6)$.

Brute force maximization of (2) has complexity $O(K^M)$, where M is the total number of impulses in the received signal; adopting ideas from the Viterbi algorithm [30], [31], algorithms with lower complexity can be found. In order to develop these algorithms, we first describe the Viterbi algorithm in a way that fits the current context. Suppose that we have to make decisions between K options at times $1, 2, 3, \ldots$, and that the process is Markov. A path to the the i-th time is a set of decisions $\mathcal{P}_i = \{\hat{D}_j; j = 1 \ldots i\}$. Let $\mathcal{P}_{i,k}$ denote the set of paths leading to $\hat{D}_i = k$. With respect to decisions at times $i+1, i+2, \ldots$ all of these paths are equivalent, that is, the optimum decisions at times $i+1, i+2, \ldots$ do not depend on which path led to $\hat{D}_i = k$ because of the Markov property. It is therefore sufficient to memorize only the *optimum* path leading to $\hat{D}_i = k$ for each k. At each stage there are K optimum paths. For the optimum path leading to $\hat{D}_i = k$ we now consider the K possible extensions to time $i+1$, and we do so for all $k \leq K$. We then again choose the K optimum paths for each of $\hat{D}_{i+1} = k$. The complexity is therefore $O(MK)$.

For timing based separations, while the timing process of each source is assumed to be Markov (i.e., a renewal process), this is generally not true for the mixture; decisions at times $i+1$,

[1] PRI = pulse repetition interval = ICI = inter-click interval = interimpulse spacing.

[2] This is not directly an optimum solution to (1), but in general ML is considered near-optimum [29], [30].

$i+2,\ldots$ do not depend only on decisions at time i. However, it is still possible to discard many paths, as outlined in the following. For a path $\mathcal{P}_i$ let $d_k(\mathcal{P}_i)$ be number of pulses since the last pulse assigned to source k. For example, if $\mathcal{P}_6 = \hat{\mathbf{D}} = (1,1,2,1,2,2)$ we would have $d_1(\mathcal{P}_6)=2$, $d_2(\mathcal{P}_6)=0$, and $d_3(\mathcal{P}_6)=\infty$. A key observation is that paths with the same K-tuple $(d_1(\mathcal{P}_i), d_2(\mathcal{P}_i), \ldots, d_K(\mathcal{P}_i))$ are equivalent with respect to future decisions, as in the Viterbi case. This is easiest seen through an example. Let's assume $\mathcal{P}_{100} = (\ldots,3,1,1,2,1,2,2)$. When impulse 101 is assigned to each of the three sources, the following term is added to the likelihood

$$\begin{aligned} &\log \mathcal{T}_1(\tau_{101}-\tau_{98}) && \text{if } D_{101}=1\\ &\log \mathcal{T}_2(\tau_{101}-\tau_{100}) && \text{if } D_{101}=2\\ &\log \mathcal{T}_3(\tau_{101}-\tau_{94}) && \text{if } D_{101}=3 \end{aligned}$$

It is clear that these terms do not depend on what happened in the part of the path before the 3, indicated by .... On the other hand, if $\mathcal{P}_{100} = (\ldots,3,2,1,1,2,1,2,2)$, we get the following modified terms

$$\begin{aligned} &\log \mathcal{T}_1(\tau_{101}-\tau_{98}) && \text{if } D_{101}=1\\ &\log \mathcal{T}_2(\tau_{101}-\tau_{100}) && \text{if } D_{101}=2\\ &\log \mathcal{T}_3(\tau_{101}-\tau_{93}) && \text{if } D_{101}=3 \end{aligned}$$

Thus, the likelihood for the decision at time 101 depends only on the tail of the path, and this is exactly characterized by $(d_1(\mathcal{P}_i), d_2(\mathcal{P}_i), \ldots, d_K(\mathcal{P}_i))$. Therefore, among paths with the same K-tuple $(d_1(\mathcal{P}_i), d_2(\mathcal{P}_i), \ldots, d_K(\mathcal{P}_i))$ we only need to memorize the *optimum* one. By definition, $d_k(\mathcal{P}_i)$ can take on $i+1$ values (i.e., $0,1,2,\ldots,i-1,\infty$), and $d_k(\mathcal{P}_i)$ must be different for different values of k. Therefore, at a time i there are $(i+1)(i)(i-1)\cdots(i+1-(K-1)) \sim i^K$ unique paths. The overall complexity (i.e., the number of paths to memorize) therefore is $O(M^{K+1})$, which is feasible to implement for moderate values of M and K. This ML algorithm is outlined in Algorithm 1, which we refer to as A1.

For large M the complexity can still be prohibitive. We therefore consider a number of simplifications to the algorithm. First we notice that many of the paths considered for the optimum solution have very low likelihood of emerging as the final optimum solution. To see this, for a path $\mathcal{P}_i$ in addition to $d_k(\mathcal{P}_i)$ define $r_k(\mathcal{P}_i)$ as the actual time since the last pulse assigned to source k. Let T_k be the time so that

$$\int_{T_K}^{\infty} \mathcal{T}_k(\tau)d\tau < \epsilon$$

for some choice of a small ϵ. If $r_k(\mathcal{P}_i) > T_k$ this means that if we extend path $\mathcal{P}_i$ with $\hat{D}_j = k$ for $j > i$, a term of at least $\log \epsilon$ will be added to the likelihood. If ϵ is small, this term is a large negative number which makes extensions of the path $\mathcal{P}$ unlikely to come out as the maximum likelihood solution; we call this a "source timeout" (STO). Thus, such paths can be eliminated from the the set of paths to keep with very little loss compared to the optimum solution – if ϵ is chosen small. This STO algorithm is outlined in Algorithm 2, which we refer to as A2.

Algorithm 1: (A1) Exact Maximum Likelihood Solution for Separation Based Solely on Timing with Known Distributions.

Given K sources and $\mathcal{T}_k$ with observations $\tau_i, i = 1,\ldots,M$

1) Initialize the assignment of τ_1 such that in the first K paths τ_1 is labeled $1,\ldots,K$. Set $i=1$.
2) For each path consider the extensions where τ_{i+1} is assigned to source $k=1,\ldots,K$
   a) Update $(d_1(\mathcal{P}_i),\ldots,d_K(\mathcal{P}_i))$ for each extended path $\mathcal{P}_{i+1}$ as follows: if the extension is $\hat{D}_{i+1}=k$ then
   $$d_k(\mathcal{P}_{i+1}) = 0$$
   $$d_j(\mathcal{P}_{i+1}) = d_j(\mathcal{P}_i)+1$$
   b) Update the likelihood of each path.
3) Put the paths with same $(d_1(\mathcal{P}_{i+1}),\ldots,d_K(\mathcal{P}_{i+1}))$ into a group.
4) In each group, eliminate all paths except the one with largest likelihood.
5) If $i+1=M$, done. Choose the most likely path as the assignment. Otherwise, increment i and go to step 2.

Algorithm 2: (A2) Maximum Likelihood Algorithm for Separation Based Solely on Timing with Known Distributions with STO Pruning.

Given K sources and $\mathcal{T}_k$ with observations $\tau_i, i = 1,\ldots,M$

1) Initialize the assignment of τ_1 such that in the first K paths τ_1 is labeled $1,\ldots,K$. Set $i=1$.
2) For each path consider the extensions where τ_{i+1} is assigned to source $k=1,\ldots,K$
   a) Update $(d_1(\mathcal{P}_i),\ldots,d_K(\mathcal{P}_i))$ and $(r_1(\mathcal{P}_i),\ldots,r_K(\mathcal{P}_i))$ for each extended path $\mathcal{P}_{i+1}$ as follows: if the extension is $\hat{D}_{i+1}=k$ then
   $$d_k(\mathcal{P}_{i+1}) = 0$$
   $$d_j(\mathcal{P}_{i+1}) = d_j(\mathcal{P}_i)+1$$
   $$r_k(\mathcal{P}_{i+1}) = 0$$
   $$r_j(\mathcal{P}_{i+1}) = r_j(\mathcal{P}_i)+\tau_{i+1}-\tau_i$$
   b) Update the likelihood of each path.
3) Put the paths with same $(d_1(\mathcal{P}_{i+1}),\ldots,d_K(\mathcal{P}_{i+1}))$ into a group.
4) Eliminate groups where for at least one k: $r_k(\mathcal{P}_{i+1}) > T_k$.
5) In each remaining group, eliminate all paths except the one with largest likelihood.
6) If $i=M$, done. Choose the most likely path as the assignment. Otherwise, increment i and go to step 2.

Algorithm 3: (A3) Simplified Algorithm for Separation Based Solely on Timing with Known Distributions with STO Pruning and a Static Number of States.

Given K sources and $\mathcal{T}_k$ with observations $\tau_i, i = 1, \ldots, M$, and a paths in memory

1) Initialize the assignment of τ_1 such that in the first K paths τ_1 is labeled $1, \ldots, K$. The assignment of the remaining $a - K$ paths is arbitrary. Set $i = 1$.
2) For each path consider the extensions where τ_{i+1} is assigned to source $k = 1, \ldots, K$
   a) Update $(r_1(\mathcal{P}_i), \ldots, r_K(\mathcal{P}_i))$ for each extended path $\mathcal{P}_{i+1}$ as follows: if the extension is $\hat{D}_{i+1} = k$ then
   $$r_k(\mathcal{P}_{i+1}) = 0$$
   $$r_j(\mathcal{P}_{i+1}) = r_j(\mathcal{P}_i) + \tau_{i+1} - \tau_i$$
   b) Update the likelihood of each path.
3) Eliminate paths where for at least one k: $r_k(\mathcal{P}_{i+1}) > T_k$.
4) For the remaining paths, keep the a paths with highest likelihood.
5) If $i = M$, done. Choose the most likely path as the assignment. Otherwise, increment i and go to step 2.

The complexity of the simplified algorithm is stochastic. Suppose that at time i there are $N(T_k)$ impulses in the interval $(\tau_i - T_k, \tau_i)$. Then the algorithm only considers paths with $d_k(\mathcal{P}_i) \leq N(T_k)$. The complexity therefore is $N(T_1)N(T_2)\ldots N(T_K)$, which is independent of M. The exact expected complexity is hard to calculate (and probably not that important), but we can do a "back of the envelope" estimate as follows. Suppose that the average impulse spacing is ρ, which depends on all the distributions $\mathcal{T}_k$. The number of impulses from source k during a time period of T_k then is approximately $\frac{T_k}{\rho}$, so the approximate total number of paths to keep is $\frac{T_1}{\rho} \cdot \frac{T_2}{\rho} \ldots \frac{T_K}{\rho}$. Assume further that $T_k \approx T$ for all k. Then the approximate complexity is $O(M(\frac{T}{\rho})^K)$, which for large M is much less than the $O(M^{K+1})$ of the optimum algorithm.

A further simplified algorithm is to keep a fixed number (or maximum number) of paths, call this number a. At each step, we maintain the paths with the largest likelihood. However, we still want to avoid any errors caused by source timeouts, so we still eliminate paths with $r_k(\mathcal{P}_i) > T_k$. This simplified STO algorithm is outlined in Algorithm 3, which we refer to as A3. Alternatively, instead of eliminating paths, we restart the algorithm whenever a source timeout is detected in the most-likely path. Another way to think of these algorithms is that we are exploring the solution space incrementally. When a source timeout error is encountered, then we have converged towards a poor solution. To get out of this mode, we forget where we came from and restart the exploration. The argument for considering only the most-likely path rests on the premise that if a source timeout occurs in a lesser path, it will probably be eliminated in further iterations. This STO restart algorithm is outlined in Algorithm 4, which we refer to as A4. In general, A4 is faster than A3 because it does not check for timeouts in every path or manage path deletion. After some experimentation, we found that $a = K^3$ works well for both versions (see Section V-A). This gives a complexity of $O(MK^3)$.

Algorithm 4: (A4) Simplified Algorithm for Separation Based Solely on Timing with Known Distributions with a Static Number of States and STO Restarts.

Given K sources and $\mathcal{T}_k$ with observations $\tau_i, i = 1, \ldots, M$, and a paths in memory

1) Initialize the assignment of τ_1 such that in the first K paths τ_1 is labeled $1, \ldots, K$. The assignment of the remaining $a - K$ paths is arbitrary. Set $i = 1$.
2) For each path consider the extensions where τ_{i+1} is assigned to source $k = 1, \ldots, K$
   a) Update $(r_1(\mathcal{P}_i), \ldots, r_K(\mathcal{P}_i))$ for each extended path $\mathcal{P}_{i+1}$ as follows: if the extension is $\hat{D}_{i+1} = k$ then
   $$r_k(\mathcal{P}_{i+1}) = 0$$
   $$r_j(\mathcal{P}_{i+1}) = r_j(\mathcal{P}_i) + \tau_{i+1} - \tau_i$$
   b) Update the likelihood of each path.
3) If in the most likely path for at least one k: $r_k(\mathcal{P}_{i+1}) > T_k$, for the next impulse, start at step 1 for the next impulse. If not, proceed to the next step.
4) For the remaining paths, keep the a paths with highest likelihood.
5) If $i = M$, done. Choose the most likely path as the assignment, and if you have restarted, concatenate all the segments together in order. Otherwise, increment i and go to step 2.

A. Theoretical Error Analysis

In this section we develop a lower bound on the error probability for separating two impulsive sources using timing only. The bound is unrelated to any specific algorithm, and can therefore be used to evaluate how close an algorithm is to optimum. It also gives insight into what kind of distributions can be separated, and how the parameters of those distributions influence separability. We derive the bound for two sources only, but the bound can be generalized to more sources.

The bound is based on a rather crude analysis. In general, there are many ways the two sources can be confused. To be able to evaluate the bound, we focus on one specific type of error event. Consider the i-th impulse of source 1, and let $j(i)$ be the nearest impulse of source 2. Let $\mathcal{A}_i$ be the event that impulses i and $j(i)$ are nearest neighbors and that the classifications of the i-th impulse and the $j(i)$-th impulse are swapped (i.e., i is assigned to 2 and j is assigned to 1). If $\mathcal{A}_i$ happens, one error for source 1 happens. Since i and $j(i)$ are nearest neighbors, the pair is disjoint from any other nearest neighbor pair. We discount all other possible errors because we are developing a lower bound. A lower bound on the error probability of

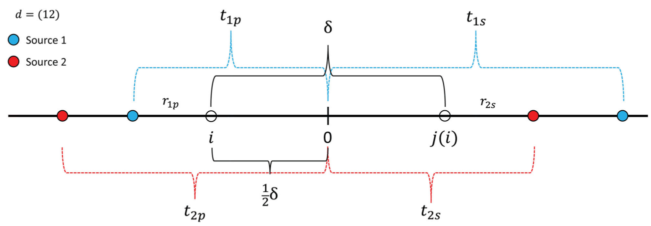


Fig. 1. Timeline illustrating the various quantities and their relations introduced for the error bound calculation for a specific source assignment. The assumption of $d = (12)$ only matters for the r_{kp}, r_{ks} distinctions. Drawing not to scale.

source 1 therefore is

$$P_{e,1} \geq \lim_{n\to\infty} E\left[\frac{1}{n}\sum_{i=1}^{n} I(\mathcal{A}_i)\right] = \lim_{n\to\infty}\frac{1}{n}\sum_{i=1}^{n} P(\mathcal{A}_i) \quad (3)$$

$$= P(\mathcal{A}_i), \quad (4)$$

where n is the number of impulses from source 1 and $I(\cdot)$ is the indicator function for event $\mathcal{A}_i$. We arrive at (4) since for large n, the $\mathcal{A}_i$ have the same probability (though they are not independent).

The probability $P(\mathcal{A}_i)$ is still difficult to calculate. However, we can find a lower bound for $P(\mathcal{A}_i)$ by letting a "genie" provide the correct decision for all impulses except i and $j(i)$. The genie also tells us that impulses i and $j(i)$ are from different sources. In the following refer to Fig. 1 for an illustration of notation. Denote $\delta = \tau_{j(i)} - \tau_i$, t_{kp} as the time-difference from the midpoint between i and $j(i)$ to the nearest prior impulse from source k, and t_{ks} as the time-difference to the nearest following impulse. Let $\mathbf{t} = (t_{1p}, t_{2p}, t_{1s}, t_{2s})$ and let $d \in \{(12), (21)\}$, where $d = (12)$ means that the first impulse is from source 1, the second from source 2. The optimum decision is now

$$\hat{d} = \operatorname{argmax} P(d|\mathbf{t}) = \operatorname{argmax} \frac{f(\mathbf{t}|d)P(d)}{f(\mathbf{t})}$$

$$= \operatorname{argmax} f(\mathbf{t}|d),$$

where $f(\mathbf{t})$ is the joint pdf of $\mathbf{t}$ obtained from the $\mathcal{T}_k$. Here we have used that $P(d) = \frac{1}{2}$ (due to symmetry). We normalize time so that the midpoint between pulses i and $j(i)$ is zero. We have

$$f(\mathbf{t}|d=(12)) = \mathcal{T}_1\left(t_{1p} - \frac{1}{2}\delta | t_{1p} > \frac{3}{2}\delta\right)$$
$$\times \mathcal{T}_1\left(t_{1s} + \frac{1}{2}\delta | t_{1s} > \frac{3}{2}\delta\right)$$
$$\times \mathcal{T}_2\left(t_{2p} + \frac{1}{2}\delta | t_{2p} > \frac{3}{2}\delta\right)$$
$$\times \mathcal{T}_2\left(t_{2s} - \frac{1}{2}\delta | t_{2s} > \frac{3}{2}\delta\right)$$
$$= \frac{\mathcal{T}_1(t_{1p} - \frac{1}{2}\delta)\mathcal{T}_1(t_{1s} + \frac{1}{2}\delta)}{P_1(t > \frac{3}{2}\delta)^2 P_2(t > \frac{3}{2}\delta)^2}$$
$$\times \mathcal{T}_2\left(t_{2p} + \frac{1}{2}\delta\right)\mathcal{T}_2\left(t_{2s} - \frac{1}{2}\delta\right)$$

$$f(\mathbf{t}|d=(21)) = \frac{\mathcal{T}_1\left(t_{1p} + \frac{1}{2}\delta\right)\mathcal{T}_1\left(t_{1s} - \frac{1}{2}\delta\right)}{P_1(t > \frac{3}{2}\delta)^2 P_2(t > \frac{3}{2}\delta)^2}$$
$$\times \mathcal{T}_2\left(t_{2p} - \frac{1}{2}\delta\right)\mathcal{T}_2\left(t_{2s} + \frac{1}{2}\delta\right), \quad (5)$$

where P_k is the probability under distribution $\mathcal{T}_k$.

If the timing process is a Poisson process, $\mathcal{T}_1$ and $\mathcal{T}_2$ are exponential, i.e., $\mathcal{T}_i(t) = \lambda_i e^{-\lambda_i t}$, so that $\mathcal{T}_1(t_{1p} - \frac{1}{2}\delta)\mathcal{T}_1(t_{1s} + \frac{1}{2}\delta) = \lambda_1^2 e^{-\lambda_1(t_{1p}+t_{1s})} = \mathcal{T}_1(t_{1p} + \frac{1}{2}\delta)\mathcal{T}_1(t_{1s} - \frac{1}{2}\delta)$ and similarly for $\mathcal{T}_2$. Therefore these two probabilities are equal and the optimum decision is achieved by randomly choosing one of the hypotheses. Since this is a lower bound, it shows

Proposition 1: Poisson processes cannot be separated purely based on timing.

In order to find the error probability, one needs to calculate the probability that $f(\mathbf{t}|d)$ is larger for the incorrect decision than for the the correct decision. This can be done for truncated Gaussians:

Theorem 2: Suppose $\mathcal{T}_i \sim \mathcal{N}_0(\mu_i, \sigma_i^2)$. A lower bound on the error probability for source 1 then is

$$P_{e,1} \geq \int_0^\infty \left(\int_x^\infty g_{X_2}(x,t)dt\right)\left(\int_{\sigma_2 x/\sigma_1}^\infty g_{X_1}(\sigma_2 x/\sigma_1, t)dt\right)$$
$$\frac{\sigma_2}{\mu_2} Q\left(x - \frac{\mu_2}{\sigma_2}\right)dx. \quad (6)$$

where

$$g_{X_i}(x,t) = \frac{1}{2\sqrt{\pi}Q\left(\frac{-\mu_i}{\sigma_i} + x\right)Q\left(\frac{-\mu_i}{\sigma_i} + 2x\right)} e^{-\frac{t^2}{4}}$$
$$\times \left(Q\left[\sqrt{2}\left(\frac{-\mu_i}{\sigma_i} + x - \frac{t}{2}\right)\right]\right.$$
$$\left. + Q\left[\sqrt{2}\left(\frac{-\mu_i}{\sigma_i} + 2x - \frac{t}{2}\right)\right] - 1\right).$$

and $Q(x) = 1 - \Phi(x)$. Note, the same holds for $P_{e,2}$; in (6) just change all $\cdot_1$ for $\cdot_2$ and vice versa.

The proof can be found in the Appendix. ■

B. Performance Comparison

In this section we compare the different algorithms from Section III, as well as the lower bound of Theorem 2. Specifically, a large number of impulses are generated for two sources with $\mathcal{T}_1 \sim \mathcal{N}_0(1, \sigma^2)$ and $\mathcal{T}_2 \sim \mathcal{N}_0(\pi, \sigma^2)$. Then the impulse trains are combined and we take the first 500 impulses as the received signal. A1, A2, A3, and A4[3] are applied to the result, and we compute the source error for finite time on the outputs as

$$P_{e,k} = \frac{\sum_i I(\hat{D}_i \neq D_i)I(D_i = k)}{\sum_i I(D_i = k)}.$$

Note, $I(\hat{D}_i \neq D_i)I(D_i = k) = 1$ i.f.f. there is an error and the correct assignment is source k; it is 0 otherwise. We adjust σ

[3]Our $\mathcal{T}_k$ are Gaussian, so we set $T_k = \mu_k + 3\sigma_k$ for the STO algorithms. This is based on the well known fact about the Gaussian that 99.7% of values will fall within in ± 3 standard deviations of the mean.

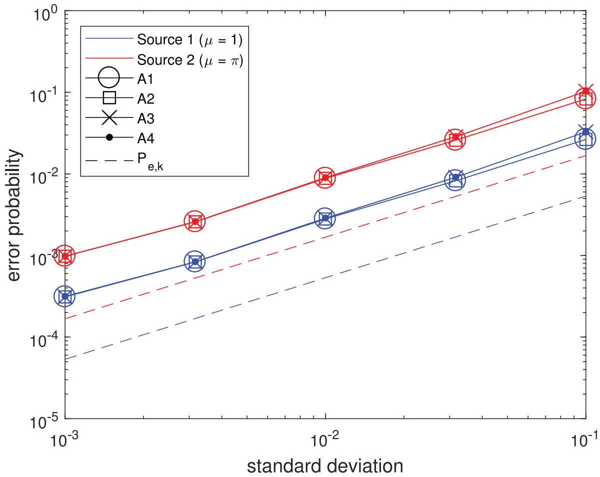


Fig. 2. For two synthetic sources with fixed mean impulse spacings (red $\mu = 1$, blue $\mu = \pi$) and identical standard deviations (indicated on x-axis), the error probability of each source from A1 (circles), A2 (squares), A3 (Xs), and A4 (dots) were averaged over 500 trials. The results are compared with the theoretical lower bound (6) shown as dashed lines.

between 10^{-3} and 10^{-1}, and repeat this process 500 times for each value. The average for the source errors over the 500 trials are computed and are compared with the lower bound; results are shown in Fig. 2. All of the proposed algorithms perform similarly: there is a constant factor of about 5 in difference between the lower bound and the algorithm performance, although, importantly, they have the same slope. We do not know if the gap is due to the bound being loose, or the fact that even the exact maximum likelihood algorithm A1 is not a direct optimum solution. As σ increases, the simplified algorithms (A3, A4) show slight degradation in performance in comparison with the "optimal" ones. Thus with some confidence we can substitute these simplified algorithms for the optimal ones for large problems in which the optimum algorithms become too complex. In this test, there is no significant difference between the two simplified algorithms.

IV. Unknown Parameters

So far we have assumed that the distributions $\mathcal{T}_k$ are known. We now extend this to the case in which we still know the total number of sources K, but $\mathcal{T}_k$ is defined by an unknown parameter vector $\boldsymbol{\theta}_k$ (e.g., $\boldsymbol{\theta}_k = [\mu_k, \sigma_k^2]$ where μ_k and σ_k^2 are the mean and variance of the distribution). We need to find $\boldsymbol{\Theta} = [\boldsymbol{\theta}_1, \boldsymbol{\theta}_2, \ldots, \boldsymbol{\theta}_K]$. However, instead of finding a single "optimum" estimate of $\boldsymbol{\Theta}$, we generate multiple estimates of $\boldsymbol{\Theta}$, and for run the pulse separation algorithms for each. The assignment that results in the largest (2) is chosen as the final output.

To generate the estimates of $\boldsymbol{\Theta}$, we can use any number of existing algorithms in literature [3], [4], [32], [33]. As an example, we consider the delta-τ histogram [3]. For a set of impulse times of interleaved impulse trains, consider every possibly pairing and compute the interimpulse spacing for each. This is an $O(M^2)$ operation. In application, the maximum spacing is usually known, making it an $O(M)$ operation. Forming a histogram of interimpulse spacings, peaks emerge at the actual interimpulse spacings for each source (and their multiples). We can extract an estimate for $\boldsymbol{\Theta}$ from these histogram peaks.

Conveniently, it can be shown that this histogram is equivalent to the integral over the histogram bins of the autocorrelation of the impulse time function [3]. As a consequence of working on a digital system, we will form the autocorrelation on a sampled impulse time function and get the same result as if we computed the autocorrelation and then integrated. Specifically, for the set of impulse times $\{\tau_i\}_{i=1}^M$, the impulse time function is

$$f(t) = \sum_i \delta(t - \tau_i),$$

a signal with an impulse at each impulse time. Its autocorrelation can be written as

$$r(s) = \sum_i \sum_j \delta((\tau_i - \tau_j) - s).$$

Each $r(s)$ is effectively a count of the number of pairs of impulse times with spacing s. If we define the start and end points of the n-th bin as (x_n, y_n), then we can write the histogram as

$$h[n] = \int_{x_n}^{y_n} r(s) ds.$$

The sampled version of $f(t)$ is defined as

$$f[n] = \int_{n\gamma}^{(n+1)\gamma} f(t),$$

where γ is the chosen sampling interval. That is, $f[n]$ is the number of impulses that lie in $[n\gamma, (n+1)\gamma]$. The corresponding autocorrelation is calculated using sums instead of integrals

$$r[s] = \sum_n f[n] f[n-s]$$

Note the n and the lag s here are sample index and not time. Multiplying the index by γ gives time. Since a binning operation was done when $f[n]$ was generated, a lag of s refers not only to a difference of $s\gamma$, but actually encompasses the range of differences $(s-1)\gamma$ to $(s+1)\gamma$. That is

$$r[n] = h[n] = \int_{(n-1)\gamma}^{(n+1)\gamma} r(s) ds.$$

In general, $r[s]$ should be computed for $s = 1, 2, \ldots, \lfloor T/\gamma \rfloor$. If a maximum possible interimpulse interval, $T_{\max}$, is known then $r[s]$ only needs to be computed up to the corresponding index. The same applies for a lower bound if a minimum possible interval is known. The choice of γ is an important topic discussed after the method is completely described.

A large histogram peak indicates that the interimpulse interval repeats many times. However, sub-harmonics (i.e., integer multiples of the actual pulse spacing) also show up as peaks, so picking the highest peaks for the μ_k is not always the best choice. Instead, we should pick peaks that are closest to the expected histogram height. In general, the height of a histogram is the total number of samples multiplied by the probability a sample falls within the range of a bin. If there is a peak in bin s, let us assume that a μ_k occurs at the center (i.e., $\mu_k = s\gamma$) and that the $\mathcal{T}_k$ are unimodal (e.g., Gaussian) and disjoint (i.e., the distributions do not overlap). The theoretical height of this peak is

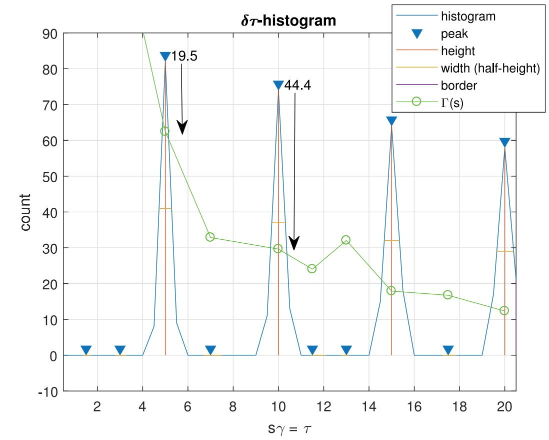
$$\Gamma(s) = \frac{T}{s\gamma} \int_{(s-1)]\gamma}^{(s+1)\gamma} \mathcal{T}_k(\tau) d\tau. \tag{7}$$

The factor before the integral is the approximate number of impulses from source with μ_k during a time period of T from III. For the integral, we make the simplifying assumption that there is no overlap between the $\mathcal{T}_k$. Overlap would lead to more spacings in this bin than suggested by the model for a single distribution per peak. That is, (7) is a conservative estimate which is preferable. To compute (7), we need to know $\mathcal{T}_k$. We already assume $\mu_k = s\gamma$ but we also need σ_k if $\mathcal{T}_k$ are Gaussian. Variance is a measure of the spread of the data, and therefore is related to the width of the peaks. We measure peak width as the distance between its boundaries at half its height[4] and use this as an estimate of σ_k (it is better to overestimate σ_k slightly than to underestimate it). Given a (μ_k, σ_k), the integral in (7) can be computed.

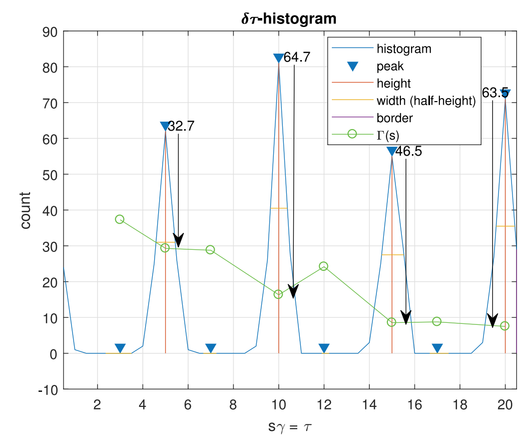
Since we pick a conservative $\Gamma(s)$, we expect that the actual μ_k should have peaks that are near but surpass it. Thus, we consider the peaks that are greater than $\Gamma(s)$, and then pick the K peaks that are closest to $\Gamma(s)$ as our estimates. This gives some protection against sub-harmonics. Sub-harmonics occur because the n-th order difference of an impulse train with interimpulse spacing μ_k will generate $T/\mu_k - n$ interimpulse spacings of $n\mu_k$. The threshold at the same point $s = \frac{n\mu_k}{\gamma}$ is $\Gamma(\frac{n\mu_k}{\gamma}) \leq \frac{T}{n\mu_k}$. Under most circumstances $\frac{T}{\mu_k} \geq 1$, so $T/\mu_k - n \gg \frac{T}{n\mu_k}$. That is, the height of a sub-harmonic peak is much greater than $\Gamma(s)$. Thus picking the peaks closest to $\Gamma(s)$ (using a simple difference measure) excludes most sub-harmonics. For example, Fig. 3(a) shows a histogram and threshold for a source with $\mu = 5$. We see the most prominent peaks occurs at μ and its sub-harmonics, but peaks at sub-harmonics are much farther away from $\Gamma(s)$ than the peak at μ is.

It is possible to have a μ_j that happens to be an integer multiple of μ_k. In this case, contribution from μ_j would add to the peak corresponding to a sub-harmonic of μ_k. This is illustrated in Fig. 3(b) which has two sources: $\mu_1 = 5$ and $\mu_2 = 10$. As before, the sub-harmonics of μ_1 are far from $\Gamma(s)$, but this time the peak at $\tau = 10$ is especially high (higher than the original sub-harmonic value) due to the addition of the impulses from the second source. To avoid ignoring the potential conflation between μ_j and the sub-harmonic of a μ_k, we pick the K farthest from $\Gamma(s)$ in addition to the K closest peaks. Thus, we have $2K$ potential peaks and extract μ_k and σ_k from their position and width, respectively. We consider every combination without replacement as a potential $\mathbf{\Theta}$. If there are $< K$ peaks that surpass $\Gamma(s)$, multiple sources might have the same μ. In this case we consider all combinations with replacement for the potential $\mathbf{\Theta}$. The entire process for generating multiple $\mathbf{\Theta}$ from the observed impulse times is outlined in Algorithm 5, which we refer to as A5.

[4]Please refer to "width (half height)" at https://www.mathworks.com/help/signal/ug/determine-peak-widths.html



(a) Histogram and threshold for a single source $\mathcal{T} \sim \mathcal{N}_0(5, 0.1^2)$.



(b) Histogram and threshold for 2 sources: $\mathcal{T}_1 \sim \mathcal{N}_0(5, 0.1^2)$ and $\mathcal{T}_2 \sim \mathcal{N}_0(10, 0.1^2)$.

Fig. 3. Example $\delta\tau$-histograms with $\gamma = 0.5$ and $T_{\max} = 20$ for $M = 100$. The black arrows indicate the distance between the peaks and $\Gamma(s)$.

If γ is too large, then all data points will fall into the same bin and there will be either one or no peaks. On the other hand, if γ is too small, data from the same distribution will be fractured into many bins and there will be many small peaks (e.g., Fig. 4). Literature suggests setting γ to the approximate jitter (i.e., σ_k) [3], [20]; this is consistent with our simulation observations. In practice, σ_k are unknown and γ is tuned to the observations. One possibility is to consider many values of γ, apply A5 and a timing separation algorithm for each of the generated $\mathbf{\Theta}$, and choose the result with largest (2) as the final output. However, this can be extremely computationally costly. An alternative is to judge a candidate histogram and adjust γ if not satisfactory. For example in Fig. 4, even though γ is too small, we can clearly see that there are 4 significant peaks of which one probably corresponds to μ (if we increase γ, we will get something like Fig. 3 a). Therefore, we suggest starting with a small γ and increasing it until the number of peaks that pass $\Gamma(s)$ is approximately

$$\sum_{k=1}^{K} \left\lfloor \frac{T_{\max}}{\hat{\mu}_k} \right\rfloor,$$

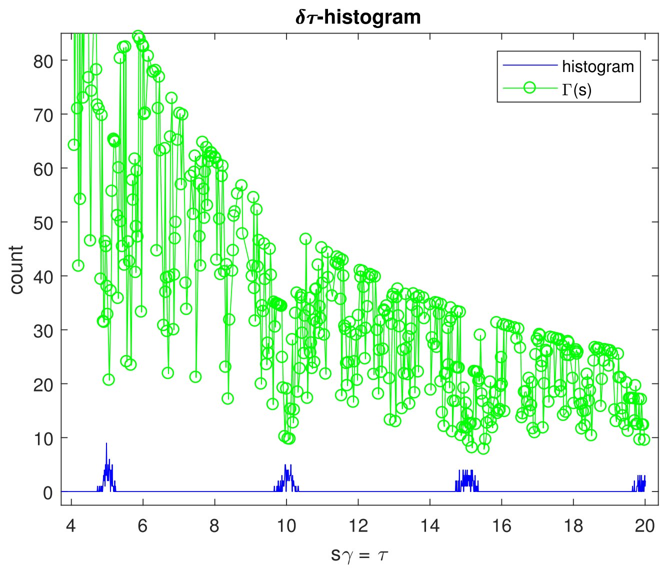


Fig. 4. $\delta\tau$-histogram with $\gamma = 0.01$ and $T_{\max} = 20$ for a single source $\mathcal{T} \sim \mathcal{N}_0(5, 0.1^2)$ and $M = 100$.

Algorithm 5: Timing Parameter Initialization Algorithm.

Given the observed impulse times $\{\tau_i\}$ for K sources and $T_{\max}$, to generate multiple $\mathbf{\Theta}$ with precision γ, do the following

1) Form a impulse time series $f[n]$ that has 1's at the impulse times and 0's everywhere else
   a) The corresponding time indexes are $\lfloor \tau_1/\gamma \rfloor$ to $\lfloor \tau_M/\gamma \rfloor$ with each point being γ apart.
2) Compute the autocorrelation, $r[s]$, of $f[n]$ for $0 < s < \lfloor \frac{T_{\max}}{\gamma} \rfloor$.
3) Find the peaks of $r[s]$ and their widths.
4) For each peak compute $\Gamma(s)$ according to (7) using the corresponding lag of each peak as μ_k and the width as σ_k. This (μ_k, σ_k^2) for the peak are the parameters collected into $\mathbf{\Theta}$ if the peak is chosen.
5) Compare the height of the peaks with $\Gamma(s)$. Considering the subsets that are greater than $\Gamma(s)$:
   a) If there are no peaks that pass, adjust γ and start again.
      i) Increase γ if there were many small peaks that did not pass
      ii) Decrease γ if there were very few peaks that did not pass (or no peaks at all)
   b) If there $< K$ peaks, then take all combinations *with replacement* of these peaks as the output.
   c) Otherwise, consider the difference $r[s] - \Gamma[s]$. Choose the K peaks that have the lowest difference and the K peaks that have the highest difference. Take all combinations *without replacement* of these $2K$ peaks as the output.

where $\hat{\mu}_k$ are the locations of peaks greater than $\Gamma(s)$ indexed in ascending order (i.e., $\hat{\mu}_1 < \hat{\mu}_2$). If $\{\sigma_k\}$ do not vary much between observations, then there is no need to tune γ every time.

As we saw, the delta-τ histogram approach is complicated by the occurrence of sub-harmonics as peaks [21]. We minimize this complication by picking peaks based on their distance from $\Gamma(s)$. Our estimates serve as input to the pulse-separation algorithms, which will likely reject the sub-harmonics. Consequently, although computational cost is increased, performance is not compromised by incorrectly picking sub-harmonics. There are other algorithms that handle sub-harmonics more directly; for example, cumulative difference (CDIF) histograms [32], sequential difference (SDIF) histograms [4], and PRI transforms [21]. However, since these are designed for periodic pulse trains, they are not automatically useful for our problem. Jitter inherent in renewal processes quickly cripples the PRI transform. [21] modified the PRI transform to make it more robust to jitter, but this relied on a heuristic algorithm that depends on many optimization parameters.[5] Similarly, the sequential search used by the CDIF and SDIF histogram methods suffer in the presence of large jitter [4], [32] (see also Fig. 9 later). On the other hand, the delta-τ histogram is robust to jitter: $\tau_{k,i} - \tau_{k,i-1} = \mu_k + \epsilon_{k,i}$, where $\mathrm{var}[\epsilon_{k,i}] = \sigma_k$, thus jitter simply widens the peaks which are used to estimate σ_k.

V. Simulated Performance Evaluation

In this section we use software to generate impulse trains according to specified truncated Gaussian distributions $\{\mathcal{T}_k\}_{k=1}^K$. We overlay the trains and then take M consecutive impulses from the result as the mixture to separate. As a performance metric we use P_e given by (1) evaluated for finite M. The source numbering is arbitrary, so we use the minimum P_e over all permutations of labelings.

A. Algorithm With Known Parameters

We first consider two synthetic sources with mean impulse spacings 1-100 (50 evenly spaced values) and standard deviations of 0.1.[6] We generate a total of 100 impulses and apply a timing separation algorithm using the actual timing parameters. We show results from A2, but for this known parameter case, all methods perform similarly. For each set of means, we compute P_e (i.e., (1)) for 500 trials and compute the average. The results are shown in Fig. 5. In general, we see that the average probability of error is low except when the source means are low (i.e., approximately $\mu_k < 20$) and similar (i.e., $\mu_i \approx \mu_j$). This is not unexpected: sources frequently emit pulses almost simultaneously when $\mu_i \approx \mu_j$, and separation is difficult when there is a near simultaneous emission from different sources.

To understand why separation is difficult for a near simultaneous emission, let A and B be pulse times that are very close.

[5] Our pulse separation algorithms can be used on the output of the modified PRI transform; our use of the delta-τ histogram is presented as one option.

[6] Our results omit units since all values are some unit of time or time^2. All values are of the same magnitude, so there is no need to specify units.

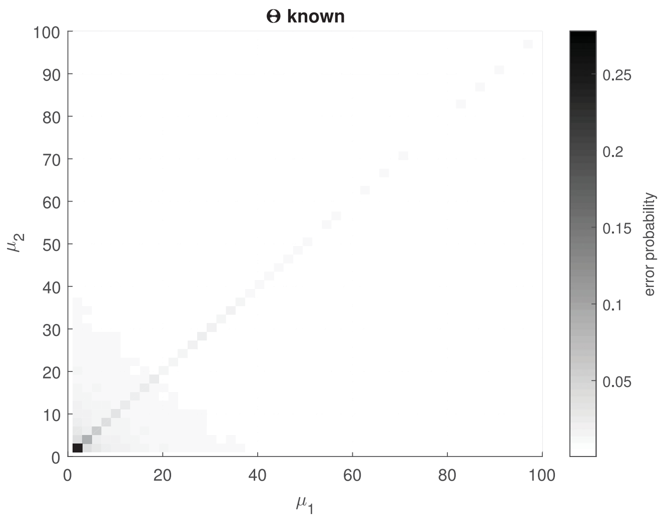


Fig. 5. Total error probability of A2 for two sources with $\sigma = 0.1$ s and μ_1, μ_2 as shown.

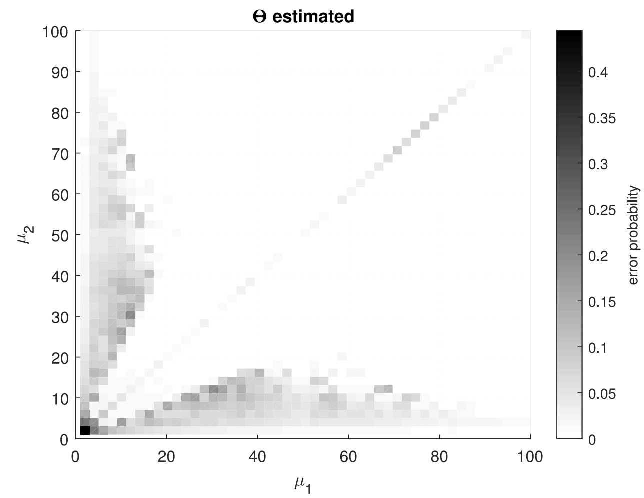
TABLE I
ERROR RATES FOR DIFFERENT VALUES OF a

a	K	K^2	K^3	K^4	K^5
$\bar{P}_e$ $(\times 10^{-2})$	0.83198	0.25730	0.25368	0.25283	0.25225
improvement (%)	-	0.5747	0.0036	0.0008	0.0006

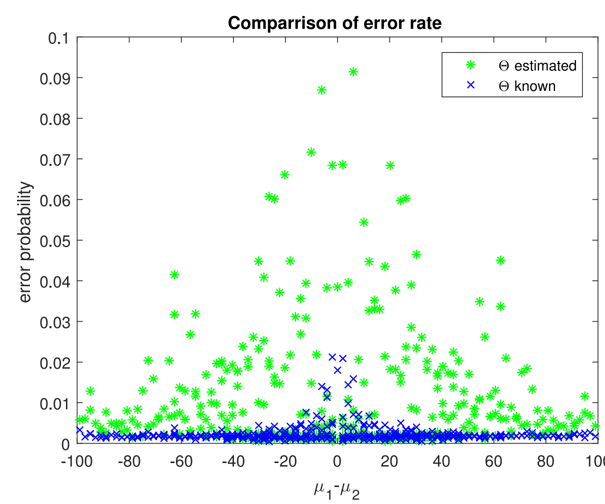
Assignments of A to source k and B to source k result in similar interimpulse intervals. The same holds for assignments of A and B to source $j \neq k$. It follows that (2) for $A \to k, B \to j$ is similar to (2) for $A \to j, B \to k$. Such assignments maybe numerically indistinguishable making the final output assignment for A and B a toss up.

Next note, when $\mu_i = \mu_j$ and $\sigma \ll \mu$, near simultaneous emissions are very likely to continue to occur after a single occurrence as the intervals will not change. The way we generate the data, there is always some chance that the first impulse times from two or more sources are almost the same whether the means are similar or not. However, this does not explain why we do not see the same amount of error for $\mu_k > 20$. In this simulation, small $|\mu_k - \sigma_k|$ corresponds to small μ_k and the distance is within regular variation. As as result, there is a higher possibility of impulse times drifting into each other. Large $|\mu_k - \sigma_k|$ reduces this possibility.

K^3 seems to be a sufficient number of paths for the approximate algorithms (A3 and A4). To demonstrate this, we repeat our simulation for $a = K, K^2, K^3, K^4, K^5$. Our performance measure is$\bar{P}_e$: the average over all $(\mu_1, \mu_2) \in [1, 100]^2$. Results are given in Table I. Performance should improve with more paths because more solutions are considered. This is indeed what we see, but the amount of improvement diminishes as K increases. Going from K^3 to K^4, improvement is $< 0.001\%$. The added computational overhead is not worth the tiny error rate reduction, and we settle on $a = K^3$. From this point on, all algorithms/results that require a use K^3 unless otherwise stated.



(a) Parameter estimation performance



(b) Parameter estimation algorithm compared with best parameter guess and actual parameter performance

Fig. 6. Total error probability of parameter estimation algorithm (using A2) for two sources.

B. Algorithms for Unknown Distribution Parameters

We next check the algorithm for unknown parameters using A5 with $\gamma = 0.5$ and $T_{\max} = 100$ to generate the candidate $\mathbf{\Theta}$. The parameters for the two synthetic sources are the same as before, but reported error rate is now averaged over 100 trials. A2 is used for the pulse separation.. We compare the error rates using parameter estimation to error rates when the actual parameters are used.

The results are summarized in Fig. 6. Fig. 6(a) gives a detailed view of the error rates for the parameter estimation algorithm. Fig. 6(b) directly compares the different cases by mapping $(\mu_1, \mu_2) \mapsto \mu_1 - \mu_2$, then averaging the error rate along like indices so results can be overlaid. Error rates for actual parameters are not shown but they are similar to Fig. 5. To be specific, A2 with known parameters gives $\bar{P}_e = 0.0025086$, while the parameter estimation algorithm gives $\bar{P}_e = 0.011957$: Performance degrades when $\mathbf{\Theta}$ is estimated from the data, but the average error rate remains small.

TABLE II
AVERAGE ERROR RATE $\times 10^{-2}$ FOR A4 WITH $K \in [2, 5]$ AND $\mu_k \in [1, 100]$, $\sigma = 0.1$

K	2	3	4	5
$\boldsymbol{\Theta}$ known	0.182	0.457	0.758	3.098
$\boldsymbol{\Theta}$ estimated	1.725	6.115	12.111	24.05

A closer analysis of Fig. 6 (a) reveals where deviations from Fig. 5 (A2 with the actual parameters) occur. First, when $\mu_1 = \mu_2$, the error rate is somewhat high. As discussed previously, our algorithms perform poorly when the source means are similar, so this is not surprising. However, with known parameters, error rates for $\mu_1 = \mu_2$ are poor only for μ_k less than about 20. In comparison, when parameters are estimated, performance is poor for some $\mu_k > 20$. A5 proposes duplicate means only if there are less than K potential peaks, but there is a possibility that a sub-harmonic or other aberrant data results in $\geq K$ peaks even when there is only a single μ_k. As mentioned in Section IV, we could check all combinations *with replacement* for all cases, but this is much more costly in terms of the number to $\boldsymbol{\Theta}$ to try.

Another difference between the cases with estimated and know parameters is the elevated error at approximately $\mu_1 < 1/2\mu_2$ for μ_1 less than about 40. This elevated error persists (but for decreasing μ_2) to $\mu_1 \approx 80$. This behavior is reflected across $\mu_1 = \mu_2$ (i.e., $1 \leftrightarrow 2$) because source numbering is arbitrary. We assume $\mu_1 < \mu_2$ and note that the same arguments apply to $\mu_1 > \mu_2$. This error is a result of using suboptimal $\boldsymbol{\Theta}$ candidates. Additive mixing of μ_i's peak with the first sub-harmonic of μ_j occurs when $\mu_i \approx 2\mu_j$ To identify μ_i nevertheless, we pick the K largest peaks above $\Gamma(s)$. This approach appears to break down in certain situations with $(\mu_1, \mu_2) < (40, 20)$. A possible explanation is that both sources have multiple sub-harmonics before $T_{\max}$ resulting in 7 or more peaks of which we pick at most 6. In this situation, our criteria may not pick the correct peak. This behavior is reduced for $\mu_1 > 40$ as the number of harmonics from the first source decreases.

To show that our algorithms are not limited to separating only two interleaved pulse trains, we consider $K \in \{2, 3, 4, 5\}$. The K means are picked at random from $[1, 100]$ and for consistency we again use $\sigma_k = 0.1$. Interleaved pulse trains are generated as before, but this time $M = 1000$. We use A4 and compare the cases where $\boldsymbol{\Theta}$ is given or estimated. P_e is calculated for each K; this is repeated 100 times (new means generated each time). The P_e are averaged over the trials and presented in Table II. Error rates increase with increasing K. In general, there is more opportunity for error when K is higher. For example, the only type of mistake that can be made with $K = 2$ is $1 \rightarrow 2$ (i.e., putting a 1 where there should be a 2 or vice versa). When $K = 3$, the types of mistakes are $1 \leftrightarrow 2$, $2 \leftrightarrow 3$, and $1 \leftrightarrow 3$. Thus it is not unexpected that the amount of error also increases with K. In the simulations, we see that the error rate roughly doubles for each increment of K whether $\boldsymbol{\Theta}$ is known or estimated. However, the error rate when $\boldsymbol{\Theta}$ is estimated is about 10 times greater than when $\boldsymbol{\Theta}$ is estimated than when $\boldsymbol{\Theta}$ is known. This may be acceptable since error is very low (0.18×10^{-2}) at $K = 2$ (which is less than the corresponding value, 0.25×10^{-2}, from Table I which used $M = 100$; more data resulted in a slight reduction in error).

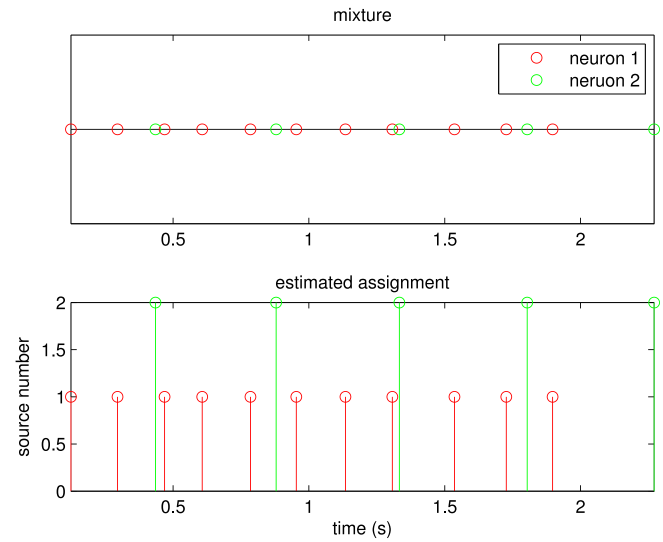


Fig. 7. (top) A simulated mixture was created using recorded action potentials. (bottom) The assignment output from A2 is exact.

We conclude that using a simple method to estimate $\boldsymbol{\Theta}$ gives good performance under most circumstances. There is room for improvement to the timing parameter estimation algorithm, possibly using some of the methods listed at the end of Section IV. However, since our main concern/contribution is our timing separation algorithm we do not pursue the parameter estimation problem further here.

VI. APPLICATIONS

In this section, we apply our algorithm to a few applications.

A. Neurons

It is often of interest to extract the activity from a single neuron. This is not possible without extracting a single neuron from the organism or using very specialized equipment. Even then, such recordings are often not practical or useful due to the specialized setup. Instead, extracellular methods can be (e.g., an EEG) in which a single electrode picks up all local activity from which individual spikes or action potentials can be identified. The process of identifying the activity from signal neurons out of the local mixture is referred to as *spike sorting* [9]. The firing rate of neurons is how information is communicated between cells. Thus, we hypothesize that the time between action potentials can be used to sort the spikes. To show that this is possible, we took two intracellular recordings (from [34]) and extracted the times of of the spikes. Then we combined the times to create a realistic mixture of extracellular recorded spike times. Specifically, the first neuron has $\theta = (0.17, 0.02^2)$, the second has $\theta = (0.45, 0.02^2)$. The timing mixture of 16 spikes spanning 2 seconds is shown in Fig. 7. We apply A2 given these parameters and achieve the correct assignment. Note that had we used

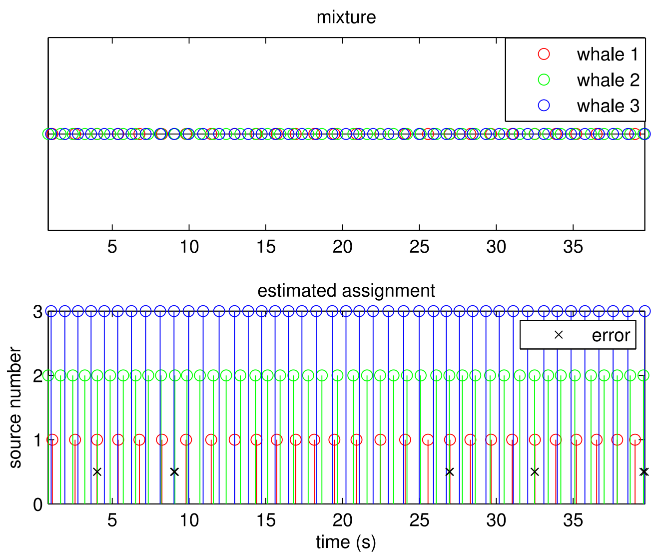


Fig. 8. (top) A simulated mixture was created using recorded sperm whale clicks. (bottom) The assignment output from A2 has 10 errors indicated with a black "x".

an actual extracellular measurement, we would have no way of knowing if the resulting classification was correct or not.

B. Sperm Whales

Clicks are high-frequency, narrowband, short-duration, quasi-impulsive signals that toothed whales use to echolocate and to orient themselves by processing returning echoes. Clicks are often produced in sequences called "click trains," with the time between clicks referred to as the ICI. ICIs can be on the order of a few microseconds to 2 s or more, and differ between species, function and situation. ICIs within a click train are generally slowly-varying [1], [2]. As such, they are a good candidate for separation by timing.

To get a realistic data set for which we had ground truth (which is difficult to obtain in reality) we tested the algorithm on recordings of a single sperm whale (from [35]) where we mixed the timings (extracted from the waveform using techniques outlined in [36]) from three different segments. Specifically, the segments with $\boldsymbol{\theta} = \{(1.3940, 0.1285^2), (0.8814, 0.0708^2), (0.9212, 0.0624^2)\}$ combined to form a mixture of 116 clicks spanning roughly 40s shown in Fig. 8. For speed and practicality, A2 with the given parameters was used in lieu of A1. The resulting assignment had 8.6% error (10 errors out of 116 clicks). Upon closer inspection, the errors occured when two clicks arriveed at nearly the same time. Recall, Section V-B discussed how near simultaneous emissions can lead to errors.

C. Radar

Unlike the previous two examples, radar pulses and timings are completely generated by humans, therefore if the right parameters are chosen, any simulation is just as good as a real world recording. Though the choice of PRI and PRI modulation will depend on the application, PRI is reported in [7] on the order from 1 μs with jitter within 8% of the PRI. We emulate this by generating 3 sources with $\mathcal{T}_k \sim \mathcal{N}_0(\mu_k, \sigma_k)$ and

$$\boldsymbol{\Theta} = \{(50, 0.80^2), (72, 0.40^2), (84, 0.04^2)\},$$

where the values are in μs and μs^2. Enough of each source was generated so that when combined, a mixture of 1000 pulses was created. Using A3 and A5 to estimate parameters with $\gamma = 1$ and $T_{\max} = 100$, there is only 1.9% error. More importantly, if we look at the source parameters according to the final output (i.e., calculating ML parameters from the assignment) we have

$$\hat{\boldsymbol{\Theta}} = \{(49.9803, 0.8654^2), (71.9770, 0.3919^2), (84.0013, 0.0560^2)\}.$$

In addition to getting a low error rate, we were able to estimate the parameters within a small margin. In this sense, the algorithm performs well.

1) Comparisons: For timing-only deinterleaving, [22] addressed problems with PRI jitter and missing pulses by using clustering,whereas [23] and [24] used the square sine wave transform and Fourier transform respectively. [25] used fuzzy adaptive resonance theory to help with this problem as well. However, without going into detail, these algorithms can be summarized as follows:

1) Get a candidate PRI (using clustering, sine wave transform, etc.).
2) Use candidate PRI in a sequential search (SS). If a sequence is found, assign it to the PRI and then remove it.

Repeat these steps until all impulses have been removed. Clustering in step 1 generates better estimates for PRI in the presence of jitter and missing pulses than other existing methods (e.g., CDIF [5] and SDIF [4]). However, the SS in step 2 was designed for constant PRI deinterleaving, while our algorithms are designed to be optimum or near optimum exactly for jittered PRI. Since our methodology does not explicitly address PRI estimation, the most relevant comparison is step 2 (SS) with our methods for known parameters. For the SS, we use the method outlined in [5] which [22] refers to in their paper. In our implementation of SS, a pulse is said to be at target time x if τ, the time of arrival of the pulse, is within $x \pm \varepsilon$. This tolerance, ε, should be a function of the jitter to prevent loss, so we let $\varepsilon = 3\sigma$. Also, because we are giving the actual PRI, a sequence with the given PRI exists, so we simply accept any sequence greater than 3 impulses rather than comparing the total (weighted) count with a threshold.

We repeat the simulation from Section III-B, with A2 and SS. This time we report P_e rather than $P_{e,k}$, extend the range of σ to 10^1 and average over 100 trials. The results are shown in Fig. 9. A2 outperforms SS throughout. Both algorithms break down at higher values of jitter.

As mentioned previously, our algorithms address pulse train separation and can be used with various parameter estimation algorithms. It might be possible to combine Liu's improved PRI estimation (step 1 above) with our algorithms (replacing step 2 above). However, this is beyond the scope of current paper.

Missed impulses may complicate the situation in practice. Our algorithms do not handle this directly, but could be adjusted to include likelihood factors for missed impulses. Without getting

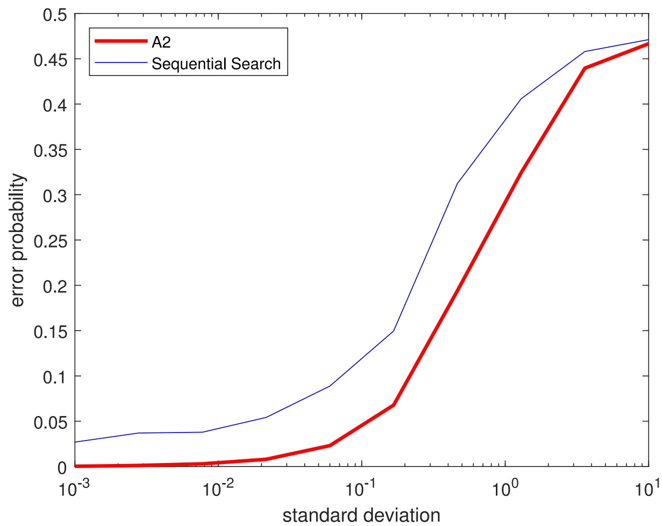


Fig. 9. Error rates averaged over 100 trials as a function of σ_k for the SS algorithm and A2 with $\mu_1 = 1, \mu_2 = \pi$.

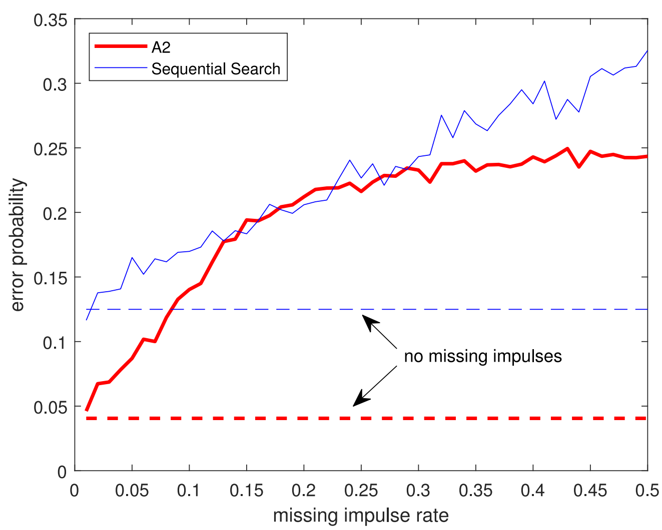


Fig. 10. Error rates over 100 trials between SS algorithm and A3 with $\mu_1 = 1, \mu_2 = \pi$ and $\sigma_k = 0.1$ for different missing impulse rates.

into this, we assess the performance of the current algorithm in the presence of missed detections. We repeat the experiment from Fig. 9 using $\sigma_k = 10^{-1}$ and a rate of missing impulses, ζ, increasing from 0 to 0.5. In each trial, $M = 100$ impulse times are generated, and ζM impulses are removed at random. Results are shown in Fig. 10. Even without modification to handle missing impulses, our algorithm matches or outperforms SS (which has built-in concessions for missing impulses).

VII. Conclusion

In this paper we have shown that separation of impulse sources is possible based solely on timing information. We developed an algorithm to classify impulses assuming knowledge of timing distributions and source number. We extended this algorithm to cases with unknown timing distribution parameters.

We presented a number of different algorithms for the same problem. Based on our experiments we can reach some limited conclusions regarding their relative performance. First, for timing separation with known parameters, our exact ML algorithms with a variable number of paths (A1 and A2) are better than the approximate algorithms with a fixed number of paths (A3 and A4). However, the difference is small, and the computational burden of the approximate algorithms are much lighter. For unknown parameters, we adapted existing parameter estimation algorithms to estimate parameter sets well enough for our algorithms to converge to a good solution under most conditions. We identified where the parameter estimation fails, and intend to modify our methods to improve performance in the future.

While our results show that pure timing separation is possible in some scenarios, timing information should be used in concert with (usually available) pulse shape for good performance on real-world data. We anticipate that doing so will improve performance and applicability and we intend to investigate this potential in a later paper.

Appendix
Proof of Theorem 2

Let's assume $d = (12)$. An error happens if (see eq. (5))

$$\mathcal{T}_1\left(t_{1p} + \frac{1}{2}\delta\right)\mathcal{T}_1\left(t_{1s} - \frac{1}{2}\delta\right)\mathcal{T}_2\left(t_{2p} - \frac{1}{2}\delta\right)\mathcal{T}_2\left(t_{2s} + \frac{1}{2}\delta\right)$$
$$> \mathcal{T}_1\left(t_{1p} - \frac{1}{2}\delta\right)\mathcal{T}_1\left(t_{1s} + \frac{1}{2}\delta\right)\mathcal{T}_2\left(t_{2p} + \frac{1}{2}\delta\right)\mathcal{T}_2\left(t_{2s} - \frac{1}{2}\delta\right).$$

In particular if we let $\mathcal{T}_i \sim \mathcal{N}_0(\mu_i, \sigma_i^2)$ an error happens if[7]

$$\begin{aligned}
&\frac{1}{\sigma_1^2}\left(t_{1p} + \frac{1}{2}\delta - \mu_1\right)^2 + \frac{1}{\sigma_1^2}\left(t_{1s} - \frac{1}{2}\delta - \mu_1\right)^2 \\
&\quad + \frac{1}{\sigma_2^2}\left(t_{2p} - \frac{1}{2}\delta - \mu_2\right)^2 + \frac{1}{\sigma_2^2}\left(t_{2s} + \frac{1}{2}\delta - \mu_2\right)^2 \\
&< \frac{1}{\sigma_1^2}\left(t_{1p} - \frac{1}{2}\delta - \mu_1\right)^2 + \frac{1}{\sigma_1^2}\left(t_{1s} + \frac{1}{2}\delta - \mu_1\right)^2 \\
&\quad + \frac{1}{\sigma_2^2}\left(t_{2p} + \frac{1}{2}\delta - \mu_2\right)^2 + \frac{1}{\sigma_2^2}\left(t_{2s} - \frac{1}{2}\delta - \mu_2\right)^2.
\end{aligned}$$

Equivalently,

$$\begin{aligned}
0 < &-\frac{1}{\sigma_1^2}(t_{1p} - \mu_1) + \frac{1}{\sigma_1^2}(t_{1s} - \mu_1) \\
&- \frac{1}{\sigma_2^2}(t_{2s} - \mu_2) + \frac{1}{\sigma_2^2}(t_{2p} - \mu_2).
\end{aligned}$$

Rewritten in terms of the inter-impulse spacing, $r_{1p} = t_{1p} - \frac{1}{2}\delta$, $r_{1s} = t_{1s} + \frac{1}{2}\delta$ etc.,

$$\begin{aligned}
\left(\frac{1}{\sigma_1^2} + \frac{1}{\sigma_2^2}\right)\delta < &-\frac{1}{\sigma_1^2}(r_{1p} - \mu_1) + \frac{1}{\sigma_1^2}(r_{1s} - \mu_1) \\
&- \frac{1}{\sigma_2^2}(r_{2s} - \mu_2) + \frac{1}{\sigma_2^2}(r_{2p} - \mu_2).
\end{aligned}$$

[7]Here the truncated Gaussian gives same result as Gaussian, as constants cancel out, and arguments are assumed positive.

Since $X > a, Y > b \Longrightarrow X + Y > a + b$, for independent random variables X, Y $P(X + Y > a + b) \geq P(X > a)P(Y > b)$, and we can therefore lower bound

$$P_{e,1} \geq P\left(-\frac{1}{\sigma_1}(r_{1p} - \mu_1) + \frac{1}{\sigma_1}(r_{1s} - \mu_1) > \frac{1}{\sigma_1}\delta\right) \times P\left(-\frac{1}{\sigma_2}(r_{2p} - \mu_2) + \frac{1}{\sigma_2}(r_{2s} - \mu_2) > \frac{1}{\sigma_2}\delta\right). \quad (8)$$

Here $r_{1p} > \delta$ by the setup, and the random variable $\frac{1}{\sigma_1}(r_{1p} - \mu_1)$ therefore has pdf (for $t > \frac{-\mu_1+\delta}{\sigma_1}$)

$$f(t) = \frac{1}{Q\left(\frac{-\mu_1+\delta}{\sigma_1}\right)\sqrt{2\pi}} \exp\left(-\frac{t^2}{2}\right)$$

Also $r_{1s} > 2\delta$ and the random variable $\frac{1}{\sigma_1}(r_{1s} - \mu_1)$ therefore has pdf (for $t > \frac{-\mu_1+2\delta}{\sigma_1}$)

$$\tilde{f}(t) = \frac{1}{Q\left(\frac{-\mu_1+2\delta}{\sigma_1}\right)\sqrt{2\pi}} \exp\left(-\frac{t^2}{2}\right)$$

The random variable $X_1 = -\frac{1}{\sigma_1}(r_{1p} - \mu_1) + \frac{1}{\sigma_1}(r_{1s} - \mu_1)$ is difference, and the pdf can be calculated using a convolution-type integral of these two pdfs, which gives[8]

$$\begin{aligned} f_{X_1}(t|\delta) &= -\frac{e^{-\frac{t^2}{4}}}{4\sqrt{\pi}Q\left(\frac{-\mu_1+\delta}{\sigma_1}\right)Q\left(\frac{-\mu_1+2\delta}{\sigma_1}\right)} \\ &\quad \times \left(\mathrm{Erf}\left[\frac{-\mu_1+\delta}{\sigma_1} - \frac{t}{2}\right] + \mathrm{Erf}\left[\frac{-\mu_1+2\delta}{\sigma_1} - \frac{t}{2}\right]\right) \\ &= -\frac{e^{-\frac{t^2}{4}}}{2\sqrt{\pi}Q\left(\frac{-\mu_1+\delta}{\sigma_1}\right)Q\left(\frac{-\mu_1+2\delta}{\sigma_1}\right)} \times \left(Q\left[\sqrt{2}\left(\frac{-\mu_1+\delta}{\sigma_1} - \frac{t}{2}\right)\right]\right. \\ &\quad \left. + Q\left[\sqrt{2}\left(\frac{-\mu_1+2\delta}{\sigma_1} - \frac{t}{2}\right)\right] - 1\right) \end{aligned}$$

where $-2\mu_1 + 3\delta \leq t$. A similar calculation can be done for the quantities related to source 2. Let

$$\begin{aligned} g_{X_i}(x,t) = &= -\frac{e^{-\frac{t^2}{4}}}{2\sqrt{\pi}Q\left(\frac{-\mu_i}{\sigma_i} + x\right)Q\left(\frac{-\mu_i}{\sigma_i} + 2x\right)} \\ &\quad \times \left(Q\left[\sqrt{2}\left(\frac{-\mu_i}{\sigma_i} + x - \frac{t}{2}\right)\right]\right. \\ &\quad \left. + Q\left[\sqrt{2}\left(\frac{-\mu_i}{\sigma_i} + 2x - \frac{t}{2}\right)\right] - 1\right) \end{aligned}$$

Given $D = \frac{\delta}{\sigma_2}$, we now write the second factor in (8) as

$$\int_D^\infty g_{x_2}(D,t)dt$$

And therefore

$$P_{e,1} \geq \int_{\frac{\sigma_2}{\sigma_1}D}^\infty g_{x_1}\left(\frac{\sigma_2}{\sigma_1}D, t\right)dt \int_D^\infty g_{x_2}(D,t)dt$$

[8]The integral was done in Mathematica.

According to renewal theory [26, Theorem 10.3.5], $D = \frac{\delta}{\sigma_2}$ has distribution

$$f_D(d) = \frac{\sigma_2}{\mu_2}Q\left(d - \frac{\mu_2}{\sigma_2}\right).$$

We can then lower bound the error probability by (6).

ACKNOWLEDGMENT

The authors thank the reviewers for their detailed feedback, which helped them to considerably improve the paper.

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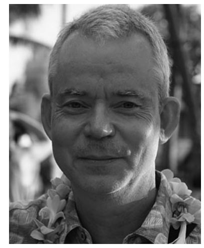

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